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Ship sailing speed optimization considering dynamic meteorological conditions

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ABSTRACT

Sailing speed optimization is a cost-effective strategy to improve ship energy efficiency and a viable way to fulfill emission reduction requirements. This study develops a novel ship sailing speed optimization method that considers dynamic meteorological conditions. We first develop an artificial neural network model for vessel fuel consumption rate (FCR) prediction based on a fusion dataset of ship noon reports and public meteorological data. Then, based on the predicted FCRs, the method repeatedly formulates a multistage graph based on the most recent forecasts, and optimal speeds for the remaining voyage are obtained until the vessel reaches the destination port. The computational efficiency of the optimization process is enhanced by progressively removing nodes without connections to successor nodes, starting from the penultimate stage. We examine the proposed method on two 11-day voyages of a dry bulk carrier. Results show that the proposed method demonstrates significant reductions in fuel consumption by 5.35% compared with a constant sailing speed scheme and by 7.34% compared with a static speed optimization model. In addition, the proposed model achieves similar fuel savings to those achieved by speed optimization based on actual meteorological conditions, enabling shipping companies to optimize ship sailing speeds in the absence of actual meteorological conditions. The proposed method can be applied to various types of vessels due to its flexibility and adaptability, making it a valuable tool for the shipping industry to reduce greenhouse gas (GHG) emissions, thereby supporting the International Maritime Organization (IMO)'s goal of reaching net-zero GHG emissions by around 2050.

1. Introduction

The International Maritime Organization (IMO) has announced in 2023 that by 2040, the total annual greenhouse gas (GHG) emissions from the maritime industry must be reduced by at least 70 % compared to 2008 levels (IMO, 2023). To achieve this goal, the IMO has implemented several regulatory initiatives aimed at reducing emissions from ships. These initiatives include the Energy Efficiency Design Index and the Energy Efficiency Operational Indicator for new and existing vessels, as well as the Ship Energy Efficiency Management Plan for all ships, promoting continuous improvement in ship energy efficiency and GHG emission reduction (IMO, 2011). In addition, starting from 2023, the operational index known as the carbon intensity indicator (CII) has been introduced, requiring Ro-Pax, cargo, and cruise ships to calculate and report their annual CII rating (IMO, 2021).

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As an active participant in the IMO, the European Union (EU) has incorporated the shipping industry into the EU Emissions Trading System (ETS), providing a market-based approach to incentivize emission reductions in the maritime sector, encouraging shipping companies to improve energy efficiency, adopt cleaner technologies, and explore alternative fuels to reduce their carbon footprint. Compliance with these regulations has become imperative for shipping companies, requiring them to improve vessel operating efficiency and thus save ship fuel and minimize environmental impacts.

To improve ship energy efficiency, the IMO has proposed various technical and operational measures. Compared to technical measures, operational measures, such as virtual arrival (Jia et al., 2017), speed optimization (IMO, 2011; Beşikçi et al., 2016b), and trim optimization (Du et al., 2019; Fan et al., 2022), are favorable for shipping companies due to their low capital investment. Among these measures, speed optimization has been widely adopted by many shipping companies, such as Maersk (Maersk, 2021) and Hafnia Limited (Hafnia Limited, 2021). This speed optimization approach relies on well-developed ship fuel consumption rate (FCR) prediction models that establish accurate correlations between FCR and various factors such as speed, draught, and meteorological conditions (Yang et al., 2020).

In recent years, numerous studies have developed sailing speed optimization models on the basis of the FCR prediction model incorporating meteorological conditions (Beşikçi et al., 2016a; Wang et al., 2018a; Yan et al., 2020). These studies assume that the meteorological conditions throughout the voyage can be obtained before departure of the ship from its origin port. The ship's sailing speeds are then optimized to achieve fuel savings using the predicted FCR from the model, along with these meteorological conditions and other sailing features as inputs. The reliability of the final optimal solution depends on the accuracy of the meteorological conditions. However, the static meteorological data used in such studies cannot reflect the ever-changing marine environment, resulting in inaccurate FCR predictions and thus undermining the effectiveness of these methods. Consequently, these methods cannot guarantee optimal ship speeds, resulting in suboptimal fuel consumption and cost savings. In addition, some studies have determined the optimal sailing speed based on real-time meteorological conditions measured by sensors installed on ships (Wang et al., 2018a; Tzortzis and Sakalis, 2021). However, the sensors can only provide the meteorological conditions surrounding the ship at the current time and not the meteorological forecasts for the remainder of the voyage. Therefore, these studies do not consider the effect of future meteorological conditions on the current speed optimization model.

On possible solution to address above issue is to use dynamically updated meteorological forecasts. In general, meteorological forecasts are considered to be accurate for relatively short time horizons (approximately two days). The accuracy of the meteorological forecasts naturally tends to degenerate as the forecast period[1] extends (Lorenz, 1965; Duan et al., 2019; Tzortzis and Sakalis, 2021). To account for this, meteorological forecast data provided by international meteorological organizations are updated at fixed time intervals. For example, the United States National Oceanic and Atmospheric Administration (NOAA) issues forecasts four times daily at 00Z,[2] 06Z, 12Z, and 18Z. Regularly updated meteorological forecasts provide more accurate and up-to-date meteorological information than static forecasts, reflecting current meteorological conditions and predicting future changes. For example, a NOAA forecast at 00Z on June 13, 2019, predicted a 2-meter temperature of 28.24 °C at $(20^\circ 00' 00'' N, 115^\circ 30' 00'' E)$ for 00Z on June 15, 2019. However, in the updated forecast at 00Z on June 15, the predicted temperature for that location and time was 29.05 °C, a difference of nearly 1 °C. Such meteorological forecasts updated in fixed time intervals enable a ship's speed profile to be re-optimized for the remainder of the voyage at the same time frequency as the updates, ensuring the reliability and accuracy of speed optimization.

This study aims to develop a novel ship sailing speed optimization model considering dynamic meteorological conditions and investigates its ability to reduce fuel consumption. First, we combine noon reports and publicly accessible meteorological information to address the data quality issue related to the meteorological information in noon reports. Based on the resulting fusion dataset, we develop and validate an artificial neural network (ANN)-based FCR prediction model. To incorporate dynamic meteorological data, we divide the ship route into segments aligned with forecast cycles.[3] The speed optimization problem is formulated as a multistage graph, where each stage includes multiple nodes representing the possible location of the vessel and is described by the remaining distance to the destination port. The weight of an edge connecting two nodes in consecutive stages corresponds to the fuel consumption required for the ship to complete the segment between the two nodes. We update the multistage graph topology based on the latest meteorological forecasts from NOAA, determining the vessel's new starting position and refreshing the weights using the FCR prediction model and recent meteorological data. The shortest path corresponding to the minimum total fuel consumption is found by utilizing the Dijkstra's algorithm, ultimately determining the optimal speed profile of the remaining voyage under the currently forecasted meteorological conditions. This updating process is repeated to generate the optimal sailing speeds for the remaining voyage segments until the vessel arrives at the destination port. The contributions of this study are list as below.

First, we develop a ship sailing speed optimization model considering dynamic environmental factors. The model continuously re-optimizes speed using recent meteorological forecasts until the ship arrives at the destination port. To incorporate dynamic meteorological forecast data, the shipping route of a voyage is divided into multiple segments based on the forecast cycle of the NOAA meteorological forecasts. Such segmentation of the shipping route allows for a more detailed consideration of the differences in meteorological conditions between segments, thereby improving the temporal resolution of the speed optimization results. The optimization problem is represented as a multistage graph to identify the optimal speeds for remaining segments. To expedite the optimization process, infeasible solutions are removed by eliminating nodes without connections to successor nodes, starting from the

[1] The forecast period represents the length of time from the initial time for which the meteorological forecast extends to the future.

[2] The "Z" in "00Z" stands for "Zulu time", which is the same as Coordinated Universal Time (UTC).

[3] The forecast cycle represents the time interval for which a new set of meteorological forecast data is generated. In this way, the time at which new meteorological forecast data are generated is exactly the time at which the ship reaches the beginning of the next segment.

penultimate stage. This approach ensures efficient and reliable speed optimization for fuel consumption reduction throughout the voyage.

Second, through the combination of noon report data with public meteorological data, the meteorological data quality of the noon report is improved. The fusion dataset obtained in this way enables us to establish a more accurate FCR prediction model than one based on using the noon report itself. This provides a more reliable foundation for subsequent ship speed optimization.

Third, we validate the speed optimization model on two 11-day voyages. Results show that the proposed method demonstrates significant reductions in fuel consumption by 5.35 % compared with a constant sailing speed scheme and by 7.34 % compared with a static speed optimization model. In addition, the proposed model achieves similar fuel savings to those achieved by speed optimization based on actual meteorological conditions, enabling shipping companies to optimize ship sailing speeds in the absence of actual meteorological conditions (e.g., when planning the sailing speed profiles of a company's next voyages).

Finally, the proposed approach can be easily implemented in practice, as it utilizes free, publicly available meteorological forecast data. Therefore, this approach is not restricted to specific ship types and routes and can be used for ship sailing speed optimization wherever data allow, providing a cost-effective and flexible solution for optimizing ship fuel consumption. Shipping companies can integrate the proposed model with existing ship energy management systems, providing an additional layer of optimization to further improve energy efficiency. The model's flexibility and adaptability will make it a valuable tool for the shipping industry to reduce GHG emissions and achieve sustainable growth.

2. Literature review

Establishing an accurate FCP model is the basis of ship sailing speed optimization (Yuan et al., 2020). Some studies have assumed that the ship FCR is proportional to the α th power of the sailing speed, where α is estimated by regression analysis (Fagerholt et al., 2010; Wang and Meng., 2012; Lai et al., 2022; Li et al., 2022a). For example, Fagerholt et al. (2010) proposed a continuous non-linear programming model to determine the optimal speed of each segment along a shipping route based on the aforementioned empirical relationship. This model was discretized and solved using the shortest-path algorithm. Subsequently, Kim et al. (2016) proposed an exact algorithm to address the same problem. Yao et al. (2012) established a relationship between the ship sailing speed and ship FCRs based on actual data, considering different sizes of container ship. Then, this relationship was embedded into an optimization model to determine the optimal refueling ports, refueling amounts, and vessel speeds on each segment of the shipping route, ultimately minimizing the overall cost associated with bunker fuel. Wang and Meng (2012) established a relationship between the vessel sailing speed and FCRs using historical operational data. Based on this relationship, they devised a mixed-integer nonlinear programming model to obtain the optimized sailing speeds for multiple container ships managed by a liner shipping network. In these studies, the ship fuel consumption was assumed to depend solely on the sailing speed. However, other factors, such as winds, currents, waves, and other meteorological conditions at sea, also affect ship fuel consumption. Neglecting these factors leads to inaccurate FCPs and thus inferior speed optimization results.

To address this limitation, many studies have recommended incorporating meteorological conditions into the construction of ship FCP models (Lee et al., 2018; Yan et al., 2020; Fan et al., 2021). Lee et al. (2018) parsed historical weather data on a given shipping route from 2012 to 2014 to determine the impacts of meteorological conditions on vessels' fuel consumption. Then, they proposed a speed optimization model aimed at minimizing ship fuel consumption and maximizing the agreement of service level. Other studies have constructed FCP models grounded in the physical principles of ship propulsion to achieve accurate estimations of FCRs for sailing speed optimization. For instance, Shao et al. (2012) adopted resistance-based model, i.e., Holtrop method, in which the added resistance due to wind and waves is considered, to estimate the fuel consumption. Then, they proposed a forward dynamic programming method which allows the ship heading and speed to change with both time and geographic positions. This method is targeted at minimizing fuel consumption during a voyage. Li et al. (2018) used a simulation model considering the impact of wind and irregular waves to calculate the bunker consumption rates of the ship's main engine. Then, they developed a bi-objective optimization model, which simultaneously minimized fuel consumption and maximized operational cost savings, to obtain the optimal sailing speeds of a fixed route. This work was later extended in Li et al. (2020) by taking into account voluntary speed loss and greenhouse gas emissions. Based on the operational data collected by sensors, Fan et al. (2021) built regression functions that related both the instantaneous fuel consumption amount and the ship speed over water with the main engine revolutions per second. Based on these two relationships, they proposed a multistage decision-making model for ship sailing speed optimization with the main engine revolutions per second as the decision variable to minimize the overall fuel consumption during the voyage.

Recently, the emergence of big data technology and artificial intelligence has paved the way for the growing adoption of machine learning algorithms in the construction of ship FCP models. Many studies have developed machine learning models that take various factors affecting FCRs as the input. For example, Tarelko and Rudzki (2020) established two ANN models to predict a ship's FCR and instantaneous speed over ground, respectively. Then, based on the predictions of the two ANN models, they proposed a bi-objective optimization model, aiming to concurrently minimize fuel consumption and maximize sailing speed. Du et al. (2019) proposed a two-phase framework with the aim of optimizing a container ship's speed and trim. In the first phase, they built ANNs to predict the vessel's FCRs based on its noon report. In the second phase, three viable measures, i.e., speed optimization, trim optimization, and speed and trim optimization, were proposed to reduce ship bunker consumption. Yan et al. (2020) trained a random forest model to predict vessels' FCRs using a dataset combining noon reports with meteorological data retrieved from the European Centre for Medium-Range Weather Forecasts (ECMWF). Then, they developed a sailing speed optimization model for the predetermined shipping route, aiming to minimize the overall fuel consumption during the voyage while guaranteeing the estimated arrival time. Based on Yan et al. (2020), Luo et al. (2023) conducted a comparative analysis to assess the effects of ensemble and deterministic meteorological forecast data on

ship speed optimization, revealing that employing ensemble meteorological forecasts for speed optimization had greater potential in reducing ship fuel consumption than did deterministic meteorological forecasts.

Some studies have taken not only the ship's sailing speed but also other factors into account, such as meteorological conditions, when constructing an FCP model. These studies demonstrate the importance of incorporating meteorological factors into the construction of accurate FCP models for optimizing ship speed profiles. These studies assumed that meteorological forecast data would be obtained *before* departure, and the vessel's FCPs based on such data would be used to plan ships' speed profiles in advance. Hence, the meteorological data and optimized sailing speeds were *determined* before ship departure and remained unchanged over the voyage. However, as the forecast period is extended, the accuracy and reliability of meteorological forecasts naturally deteriorate due to the chaotic nature of the ocean and atmosphere (Yousuf et al., 2019; Tzortzis and Sakalis, 2021). Consequently, FCP results based on pre-obtained meteorological forecasts suffer from deteriorating accuracy, making the results of speed optimization models unreliable, especially for long-distance voyages.

In some studies, to tackle the issue of deteriorated accuracy of pre-obtained meteorological forecasts, the real-time meteorological information captured by sensors installed on ship can be used in speed optimization model. For example, Tzortzis and Sakalis (2021) divided the full-time horizon into smaller time regions during which the weather conditions provided by sensors are considered as accurate, and successive limited time horizon sub-problems are solved to determine the optimal ship speed, aiming at minimizing the overall fuel consumption for a given voyage. Wang et al. (2018a) adopted a novel dynamic optimization method to determine the optimal sailing speeds under the specific real-time updated environmental factors provided by sensors at different time steps. The sensors can only provide the meteorological conditions surrounding the ship at the current time and not the meteorological forecasts for the remainder of the voyage. Therefore, these studies do not consider the effect of future meteorological conditions on the current speed optimization model.

To address the shortcomings of static meteorological forecast data and shortsighted sensor data, continuously updated meteorological forecasts can be utilized to characterize the continuously changing marine environment. Specifically, meteorological forecasts are updated according to a fixed cycle, implying that up-to-date meteorological forecast data can be obtained for sailing speed optimization during a ship's voyage. By using such information, we can update the predictions of FCP models and obtain more reliable speed optimization results.

A viable solution is to adopt the idea of isochrones (Lin et al., 2013; Roh, 2013; Zis et al., 2020), which indicates the contour lines that the ship can reach with equal sailing time. In the conventional isochrones-based weather routing method (Roh, 2013), a set of possible sailing directions started from the departure point is determined by discrete heading angles in the first time stage. Then, the first isochrone is constructed by connecting the waypoints in each direction that can be reached within a given time limit and the same desired sailing speed. These waypoints are regarded as new start points to generate the waypoints and the isochrone in the next stage. This process is repeated until the destination point is connected by the last isochrone. For any two adjacent isochrones, the fuel consumption on all feasible connections is calculated based on the meteorological forecast obtained before departure. The optimal route is recognized as the one having the minimum total fuel consumption. Another realization of isochrones-based weather routing can be found in Lin et al. (2013). Notably, their construction of isochrones is based on the discretization of both sailing speed and the headings, which is different from Roh (2013). The optimization objective is to find the route with a minimum speed loss. However, these studies (Roh, 2013; Lin et al., 2013) assume that the meteorological forecast obtained before departure can accurately describe the meteorological conditions for the entire journey in the coming days, without obtaining further updated meteorological forecasts during the voyage.

In this study, the idea of isochrone-based weather routing is extended. We divide the ship route into segments aligned with forecast cycles. Instead of discretizing heading angles in Roh (2013) or discretizing sailing speed in Lin (2013), we innovatively discretize the range of the remaining distance to the destination port to create nodes in the next stage. This process can be regarded as generating the next isochrone. Moreover, in contrast to these isochrone methods (Lin et al., 2013; Roh, 2013), we further incorporate the constantly updated meteorological forecast in the construction of the isochrone. Specifically, we formulate the speed optimization problem as a multistage graph based on the segmentation, where each stage includes multiple nodes representing the possible location of the vessel and is described by the remaining distance to the destination port. The weight of an edge connecting two nodes in consecutive stages corresponds to the fuel consumption required for the ship to complete the segment between the two nodes. We update the multistage graph topology based on the latest meteorological forecasts from NOAA, determining the vessel's new starting position and refreshing the weights using an FCP model and recent meteorological data. The shortest path corresponding to the minimum total fuel consumption is found by utilizing Dijkstra's algorithm, ultimately determining the optimal speed profile of the remaining voyage under the currently forecasted meteorological conditions. This updating process is repeated to generate the optimal speed profile for the remaining voyage segments until the vessel arrives at the destination port. The first speed value in the speed profile obtained in each run forms the ship's optimal sailing speed profile to accomplish the voyage.

3. Data description

3.1. Ship noon report

Ship's noon report, which is typically prepared by the ship's captain and sent to onshore officers daily (usually at noon), is an essential document that provides a comprehensive overview of the ship's sailing conditions and the surrounding meteorological conditions (Yan et al., 2020). The sailing conditions provided by the report generally include the recording time, geographical coordinates, distance traveled since the previous report, steaming time since the previous report, average sailing speed since the previous

report, sailing condition (laden or ballast), and average propeller revolutions per minute. The surrounding meteorological conditions are recorded manually once per day based on the ship crew's readings from onboard instrumentation or visual observations. This section of the report includes information on the wind force and direction, sea swell height and direction, and sea current depth and direction. The extent of information contained in this report is limited and may vary between reports. Nonetheless, noon reports are the primary source of information in research related to predicting and optimizing ship fuel consumption.

In this study, the noon reports are obtained from a Capesize dry bulk carrier. The reports cover the time span from February 13, 2019 to August 3, 2020, consisting of a total of 331 records. To validate the effectiveness of the proposed sailing speed optimization model that incorporates dynamic meteorological conditions, two voyages are randomly selected from the dataset for computational experiments. As these selected noon report data are not involved in the training of the FCP model, they are also used as the test set to assess the FCP model's predictive performance. This enables us to maximize the utilization of limited data. As a result, the training set contains 307 (92.7 %) samples, and the test set contains 24 (7.3 %) samples.

3.2. ECMWF reanalysis v5 (ERA5)

To mitigate the inaccuracy and limitedness of the meteorological data provided by the noon reports, we seek to improve the meteorological data quality by leveraging more reliable external sources. In this study we use the ERA5 dataset, a reanalysis dataset that is widely considered the most comprehensive and accurate representation of historical weather and climate conditions. This dataset offers hourly estimations of various meteorological variables from 1979 to the current day, with a horizontal resolution of $0.25^{\circ} \times 0.25^{\circ}$ for the atmosphere and $0.5^{\circ} \times 0.5^{\circ}$ for ocean waves (ECMWF, 2018). Therefore, we regard the historical meteorological conditions from ERA5 as the ground truth. In many studies on fuel consumption prediction, meteorological data provided by ERA5 were incorporated into noon reports to generate an augmented dataset, which was used to train an FCP model with satisfactory predictive performance (Yan et al., 2020; Li et al., 2022b; Luo et al., 2023). In this study, six variables from the ERA5 data are retrieved: mean wave direction, significant height of combined wind waves and swell, wind speed, wind direction, forecast surface roughness, and 2-meter temperature. These meteorological variables are combined with the sailing features obtained from the noon reports to develop the FCP model, leading to more reliable and accurate predictions of fuel consumption. The features used for this prediction task are shown in Table 1. The absolute wind direction and absolute mean wave direction from the ERA5 data are converted to relative directions with respect to the ship's true heading. We assume that the ship sails along the rhumb line. Therefore, the vessel's true heading is calculated by the following equations (Veness et al., 2010):

$$\Delta\lambda = \begin{cases} \lambda_2 - \lambda_1 - 2\pi, \ if \ \lambda_2 - \lambda_1 > \pi \\ \lambda_2 - \lambda_1 + 2\pi, \ if \ \lambda_2 - \lambda_1 < -\pi \\ \lambda_2 - \lambda_1, otherwise \end{cases} \tag{1}$$

$$\Delta\psi = \ln\left(\frac{\tan\left(\frac{\pi}{4} + \frac{\varphi_2}{2}\right)}{\tan\left(\frac{\pi}{4} + \frac{\varphi_1}{2}\right)}\right) \tag{2}$$

Table 1
Data overview.

Feature/target	Description	Min value	Max value	Mean value
Prediction target				
Hourly fuel consumption rate of the main engine	The average fuel consumption rate of the ship's main engine in unit metric ton (mt)/hour.	0.434	2.184	1.343
Features obtained from ship noon report				
Average sailing speed	The average ship sailing speed since the last report in unit knot.	6.730	14.920	11.068
Sailing condition	The voyage is ballast, if the vessel does not carry cargo; The voyage is laden, if the vessel carries cargo.	/	/	/
Draught	The vertical distance in unit meters from the waterline to the hull's bottom.	8.600	18.700	13.885
Features obtained from ECMWF				
Relative wind direction	The wind direction at the height of 10 m and relative to the vessel's true heading in unit degrees.	0.150	179.675	93.425
Wind speed	The wind speed at the height of 10 m in unit meters per second.	0.421	15.178	5.432
Mean wave period	The average time it takes for two successive wind-sea wave crests to pass through a fixed position on the sea surface in unit second.	2.395	13.380	7.138
Significant height of combined wind waves and swell	The vertical distance in meters between the wave crest and the wave trough, considering both the wind-sea wave and the swell.	0.150	3.975	1.435
Relative mean wave direction	The mean direction of sea surface waves and relative to the vessel's true heading in unit degrees.	0.769	179.494	87.659
Forecast surface roughness	The length of aerodynamic roughness in meters, serving as a measure of surface resistance.	2.420E-05	7.230E-04	0.061
2-meter temperature	The air temperature at 2 m above the earth's surface, expressed in unit °C.	2.286	29.311	25.039

$$\theta = \left(\frac{\arctan \frac{\Delta\psi}{\Delta\lambda}}{\pi} \times 180^\circ + 360^\circ \right) \% 360^\circ, \tag{3}$$

where λ_1 and φ_1 are the longitude and latitude of the origin port expressed in radians, respectively; λ_2 and φ_2 are the longitude and latitude of the destination port expressed in radians, respectively; $\Delta\lambda$ is the difference in longitude between the origin and destination ports; $\Delta\psi$ is the difference between the latitude of the origin and destination ports; $\ln(\bullet)$ represents the natural logarithm; and θ is the vessel's true heading between the origin and destination ports when the vessel travels along the rhumb line and ranges between 0° and 360°.

3.3. NOAA meteorological forecast data

In practical sailing speed planning, shipowners can leverage meteorological forecasts to gain advance knowledge of the meteorological conditions that will be encountered by the ship during the voyage. Armed with this information, shipowners can plan the ship's sailing speed before departure with the goal of minimizing fuel consumption during the voyage. In this study, the meteorological forecast data are obtained from the Global Ensemble Forecast System (GEFS) developed by the NOAA (NOAA, 2017). The GEFS generates a range of 21 meteorological forecasts, consisting of a "control member" forecast that is based on the most accurate estimate of the current meteorological conditions and 20 "perturbed member" forecasts that are generated from the perturbed initial meteorological conditions(WMO, 2012). The control member forecast represents the most likely scenario, while the perturbed member forecasts provide a measure of forecast uncertainty. The NOAA issues meteorological forecast data four times a day, with the timestamps of 00Z, 06Z, 12Z, and 18Z. The issued dataset produces forecasts every six hours from the initial time, which refers to the starting point from which the forecast data are generated. The forecast data extend up to 384 h in the future (i.e., 16 days) from the initial time and cover various parameters such as temperature, wind speed, precipitation, and atmospheric pressure with a horizontal resolution of $0.5^\circ \times 0.5^\circ$. As this study does not focus on the impact of meteorological forecast uncertainty on sailing speed planning, we use solely the control member forecast as a description of future meteorological conditions. In this study, three variables from the NOAA meteorological forecast data are retrieved, namely the 2-meter temperature, the wind speed, and the wind direction.[4] These three variables share identical names in the NOAA and ECMWF forecasts. Similarly, the absolute wind directions recorded in the NOAA data are converted to the relative direction with respect to the vessel's true heading.

4. Development of ANN models for predicting ship fuel consumption

ANNs are computational models inspired by the neural networks of the human brain. These networks are composed of interconnected nodes, often referred to as "neurons" (Hassoun et al., 1995; Hastie et al., 2009). Each neuron in an ANN receives multiple inputs, which it processes to compute an output. The computation typically involves calculating a weighted sum of the inputs and inputting the sum to an activation function. Let us denote the inputs as a vector $\boldsymbol{x} = [x_1, \cdots, x_n]^{\mathrm{T}}$ and the corresponding weights as $\boldsymbol{w} = [w_1, \cdots, w_n]^{\mathrm{T}}$. The operation of a neuron can be expressed as:

$$h = f\left(\boldsymbol{w}^{\mathrm{T}}\boldsymbol{x} + b\right), \tag{4}$$

where $\boldsymbol{w}^{\mathrm{T}}\boldsymbol{x}$ denotes the weighted sum of the inputs; b is the bias term, which is also a parameter of the neuron; and f is the activation function, which introduces non-linearity into the output of a neuron, enabling the network to learn from more complex patterns. Common choices for f include sigmoid, tanh, and rectified linear unit (ReLU).

An ANN can be represented as a directed acyclic graph, where each node in the graph corresponds to a neuron and each edge represents a weighted connection between neurons. In general, an ANN is typically organized into layers, including the input, hidden, and output layers. The input layer of the network is composed of nodes representing the input values, and the output layer is composed of nodes representing the corresponding output value(s). In between, there may be one or more hidden layers, each containing one or more neurons. The number of hidden layers, as well as the number of neurons in each hidden layer, can vary with the complexity of the prediction task. To illustrate the structure of an ANN, let us consider a simple three-layer network that is used to predict the FCR, as shown in Fig. 1.

Let $z_t (t = 1, \cdots, T)$ denote the weighted sum of the inputs to all neurons in the hidden layer. z_t can be calculated by

$$z_t = \sum_{j=1}^{10} w_{jt} x_j = \boldsymbol{w}_t^{\mathrm{T}} \boldsymbol{x}, t = 1, \ldots, T, \tag{5}$$

where w_{jt} is the weight of connection between the j th neuron in the input layer and the t th neuron in the hidden layer. The output of neurons in the hidden layer, denoted by $g_t (t = 1, \cdots, T)$, is calculated by

[4] The NOAA ensemble forecast data only provide atmospheric weather forecast information, and do not include oceanographic meteorological data, such as wave height or sea surface temperature.

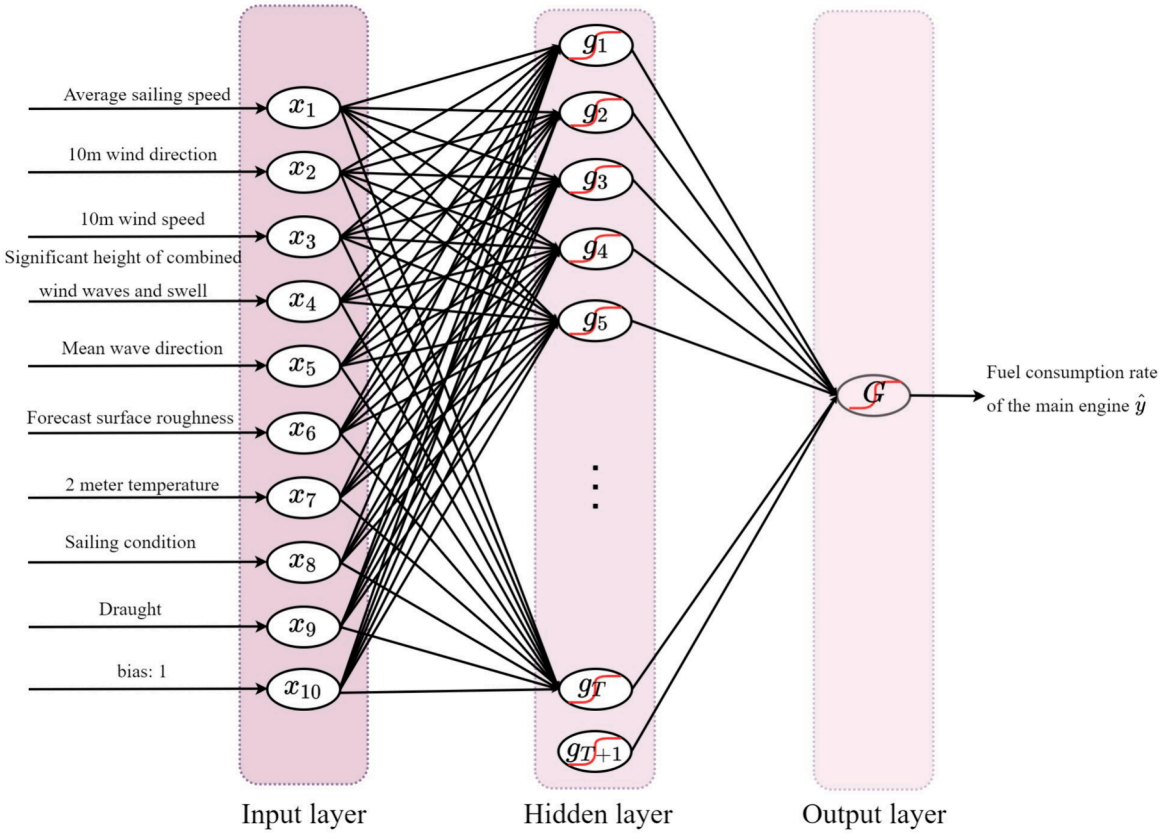


Fig. 1. The structure of a three-layer ANN model.

$$g_t = f(z_t),\ \ t = 1, \cdots, T \tag{6}$$

$$g_{T+1} = 1. \tag{7}$$

$g_t(t = 1, \cdots, T)$ and g_{T+1} are then considered as the inputs to the output layer. The output of the neurons in the output layer, denoted by G, is calculated by

$$G = f\left(\sum_{t=1}^{T+1} \partial_t g_t\right), \tag{8}$$

where ∂_t is the weight of connection between the t th neuron in the hidden layer and the output layer. The predicted value is the output of the neurons in the output layer, which is calculated by

$$\widehat{y} = f\left(\sum_{t=1}^{T+1} \partial_t g_t\right) = f\left(\sum_{t=1}^{T+1} \partial_t f(\boldsymbol{w}_t^{\mathrm{T}} \boldsymbol{x})\right). \tag{9}$$

Once we have computed the output $\widehat{y}$, we can compare it to the true output y and use a loss function to measure how well the ANN network performs. One common loss function for regression tasks is half of the mean squared error, which is calculated as follows:

$$L(y, \widehat{y}) = \frac{1}{2n} \sum_{i=1}^{n} (y_i - \widehat{y}_i)^2. \tag{10}$$

To learn the weights and biases of the network, including $w_{jt}(j = 1, \cdots, 10; t = 1, \cdots, T)$ and $\partial_t(t = 1, \cdots, T+1)$, we use the back-propagation algorithm to compute the gradients of the loss function with respect to the network parameters. The gradients are then used to update the weights and biases in the direction that minimizes the loss function. This process is typically repeated many times using optimization algorithms such as adaptive moment estimation (Kingma and Ba, 2014) or stochastic gradient descent, which adjust the weights and biases iteratively based on the computed gradients. Once we have obtained the weights and biases, we can construct the FCP model, denoted by $f^{ANN}(v, \mathbf{w}, \mathbf{u})$. In this model, $v = x_1$ is the sailing speed, $\mathbf{w} = (x_2, \cdots, x_7)$ represents the meteorological conditions, and $\mathbf{u} = (x_8, x_9)$ represents the draught and sailing conditions (laden or ballast).

To evaluate the predictive performance of $f^{ANN}(\bullet)$ on a dataset, three common measures for evaluating regressor performance are adopted, namely mean squared error (MSE), mean absolute error (MAE), and mean absolute percentage error (MAPE). Using $\bar{n}$ to represent the total number of samples in the dataset, the formulas for the MSE, MAE, and MAPE are as follows:

$$MSE = \frac{1}{\bar{n}} \sum_{i=1}^{\bar{n}} \left(f^{ANN}(\boldsymbol{x}_i) - y_i\right)^2 \tag{11}$$

$$MAE = \frac{1}{\bar{n}} \sum_{i=1}^{\bar{n}} \left|f^{ANN}(\boldsymbol{x}_i) - y_i\right| \tag{12}$$

$$MAPE = \frac{100\%}{\bar{n}} \sum_{i=1}^{\bar{n}} \left|\frac{f^{ANN}(\boldsymbol{x}_i) - y_i}{y_i}\right|. \tag{13}$$

5. Development of ship sailing speed optimization model

5.1. Problem statement

The notation used in this study is summarized in (See Table 2).

To illustrate how to incorporate the continuously updated meteorological forecast data into the speed optimization problem, we consider a scenario in which a ship travels along a predetermined shipping route from origin port O to destination port D, as shown in Fig. 2. The route can be divided into $n = \lfloor (T_{\max} - (T - \Delta t))/T \rfloor + 1$ segments according to the forecast cycle of the meteorological forecasts, where $T_{\max}$ is the maximum allowable traveling time from port O to port D, T is the meteorological forecast cycle, and $\Delta t = T^1_{\text{start}} - T^1_{\text{nearest}}$ (T^1_{nearest} is the timestamp of the latest historical meteorological forecast before the departure time of the first run). This implies that the sailing time on the first segment is $T - \Delta t$, that on the final segment is $T_{\max} - (n-1) \times T + \Delta t$, and that on the j th ($j \in \{2, 3, \cdots, n-1\}$) segment is T. This segmentation mechanism ensures that the total sailing time is no larger than the maximum allowable sailing time, which provides a potential time constraint for multistage graph based optimization, as will be seen in Section 5.2.2. Within a given segment, it is assumed that the meteorological conditions, which are represented by the latest meteorological forecast at the beginning of the segment, remain consistent. Additionally, the sailing speed is constrained within the range $[v_{\min}, v_{\max}]$, where $v_{\min} = 8.0$ and $v_{\max} = 18.0$ knots, and we assume that the sailing speed remains unchanged during any one segment of the voyage. For each segment, the optimal speed is determined through a speed optimization model based on the most recent meteorological forecasts. The ship then sails at this optimal speed until the beginning of the next segment, while a new run of the speed optimization model is performed using the latest meteorological forecast data. This process is repeated to generate the optimal sailing speeds for remaining segments using the most up-to-date meteorological forecasts until the ship arrives at its destination port, and the first speed value in the speed profile obtained in each run forms the most up-to-date optimal sailing speed profile to complete the voyage.

Specifically, in the first run, the ship is located at port O, as shown in Fig. 2, and the meteorological conditions over the whole voyage are obtained in advance from the latest meteorological forecast. If the ship departs from port O at 01Z on July 14, the latest meteorological forecast is the one updated at 00Z on July 14. Therefore, the meteorological conditions for each segment are those at the beginning of the segment, denoted by $\mathbf{w}_{00Z}^{\rho\times(j-1)}(O^j)(j \in S)$, and are obtained from the meteorological forecast with initial time 00Z and forecast time step ρ. Based on the obtained meteorological conditions, the optimal speeds at all the remaining segments can be determined by the sailing speed optimization model. Then, the ship navigates at the speed v_1^1, which is the optimal speed for segment 1 obtained by solving the sailing speed optimization model in the first run. When the ship arrives at the beginning of segment 2, the meteorological forecasts for the remaining voyage are updated as $\mathbf{w}_{06Z}^{\rho\times(j-1)}(O^j)(j \in \{2, \cdots, n\})$. Then, the optimal speed profile for the remaining $n-1$ segments can be re-solved by rerunning the sailing speed optimization model with the updated meteorological information, and the ship navigates with the re-optimized speed v_2^2 along segment 2. The meteorological forecast in the n th run is given by $\mathbf{w}^0_{(c\times\rho)Z}(O^n)$, where $c = (n-1) - \left\lfloor \frac{n-1}{4} \right\rfloor \times 4$. This speed optimization is repeated at most n times until the ship arrives at port D. The pseudo-code of the sailing speed optimization considering the constantly updated meteorological forecast is presented in Algorithm 1 of Appendix B.

5.2. Multistage graph based ship sailing speed optimization

In Section 5.1, we illustrated the process of generating the optimized sailing speed profile for the remaining voyage using the constantly updated meteorological forecast data. In general, in the k th run of optimization, the remaining route is divided into N^k segments, and the speed value of each segment is limited within the range $[8.0, 18.0]$. Assuming that the speed takes discretized values within this range at intervals of 0.1, there are 101 possible speed values. Therefore, for the speed optimization problem in the k th run, there are 101^{N^k} speed profiles. It is computationally intractable to compare the ship fuel consumptions induced by all these speed profiles. To make this problem solvable, we propose a multistage graph-based solution for the sailing speed optimization problem and

Table 2
Notation of the ship sailing speed optimization model.

Sets and indices		
n	The total number of segments over a voyage	
S	The set of all segments,$S = \{1, \cdots, n\}$	
j	Index of a segment,$j \in S$	
k	The k th execution of the ship sailing speed optimization model	
i	Index of a stage	
O^j	The beginning of segment j	
Parameters	**Symbols**	**Interpretation**
Parameters for meteorological forecasts	T	Forecast cycle of meteorological forecasts, which refers to the time interval at which a numerical weather prediction (NWP) model is run to generate a new set of forecast data.
	ρ	Forecast time step of meteorological forecasts, which refers to the time interval between successive forecast outputs generated by an NWP model during a forecast cycle.
	$\mathbf{w}_{initial_time}^{time_step}(O^j)$	The meteorological conditions at the beginning of each segment, where $time_step = \{0, \rho, \cdots, 64\rho\}$, $initial_time = \{00Z, 06Z, 12Z, 18Z\}$, and $j \in S$.
Parameters for the sailing speed optimization	L	The total length from the origin port to the destination port over a voyage
	$T_{\max}$	The maximum allowable traveling time over a voyage
	$v_{\min}$	The minimum allowable sailing speed along a voyage
	$v_{\max}$	The maximum allowable sailing speed along a voyage
	ζ	Distance interval, which is used to discretize the range of the remaining distance to the destination port represented by the nodes in each stage in the multistage graph.
	L^k	The remaining distance to be traveled in the k th run
	T_{start}^k	The departure time of a ship in the k th run of the speed optimization model. When k equals 1, T_{start}^1 is the ship's departure time from port O.
	T^k	The remaining maximum allowable traveling time in the k th run
	N^k	The total number of segments in the k th run
	$f^{\text{ANN}}(v, \mathbf{w}, \mathbf{u})$	The ship's predicted FCRs derived from the ANN model when the vessel sailing speed is v, meteorological conditions are $\mathbf{w}$, and draught and sailing conditions (laden or ballast) are $\mathbf{u}$.
	λ_1	The longitude of the departure port
	φ_1	The latitude of the departure port
	θ	The ship's true heading when the ship is at the departure port.
Decision variables		
v_j^k	The optimal speed on segment j at the k th run of sailing speed optimization model	
t_j^k	The sailing time on segment j at the k th run of sailing speed optimization model	

remove paths representing infeasible solutions from the multistage graph, further reducing the computational burden. In the following subsections, we first introduce the concept of a multistage graph and then provide a detailed explanation of how to construct such a graph to solve the sailing speed optimization problem.

5.2.1. *Introduction of multistage graph*

A multistage graph $G = (V, E)$ is a directed and weighted graph with vertices V and edges E. The vertices are divided into $n(n \geq 2)$ disjoint subsets, $V = \{V_1, \cdots, V_n\}$, where each subset V_i denotes the set of nodes for a stage of the multistage graph. V_1 only contains one node, called the source, and V_n only contains one node, called the sink, while the remaining subsets $V_i(i = 2, \cdots, n-1)$ represent the intermediate nodes that connect the source and sink nodes and may include more than one node. The values of nodes represent the nodes' attributes of interest. Edges exist only between adjacent stages in the graph, and thus the set of edges can be represented by $E = \{(u, v) | u \in V_i, v \in V_{i+1}, 1 \leq i \leq n-1\}$. Each edge has an associated weight (cost) $c(u, v)$.

In a multistage graph, a path is a sequence of connected nodes that start from the source and end at the sink. From such a graph, we can find the path from the source node to the sink node with the lowest total weight, which is also referred to as the shortest path. Fig. 3

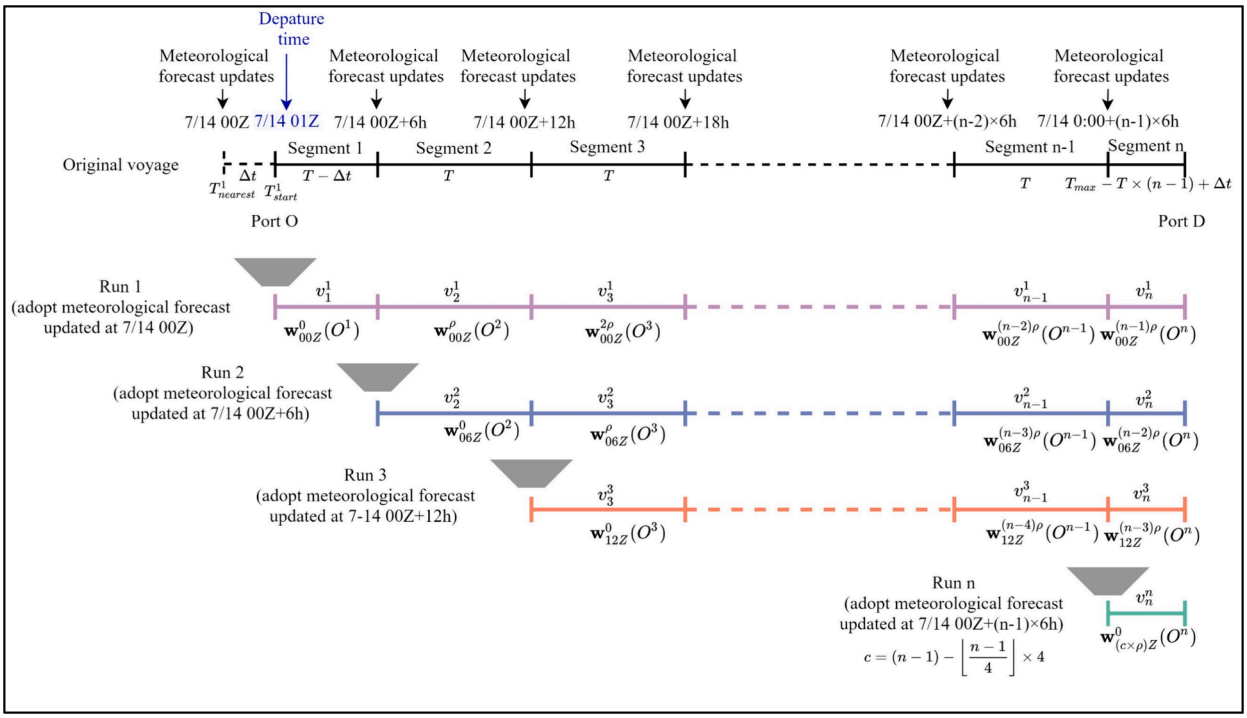


Fig. 2. An illustration of sailing speed optimization considering the constantly updated meteorological forecasts.

shows a four-stage graph, in which the numbers on the edges represent weight values and the values of nodes represent the index. The shortest path from the source to the sink on this four-stage graph is 0→1→4→6, with a total weight of 14.

5.2.2. Construction of the multistage graph for sailing speed optimization problem

The multistage graph introduced in Section 5.2.1 can be used to address the sailing speed optimization problem proposed in Section 5.1. In this context, each stage of the graph represents the location of the vessel (i.e., waypoint) at each update time of the meteorological forecast data. Each stage contains a set of nodes, and each node represents a possible location of the vessel. For each node, the distance to the sink node (i.e., the remaining distance to be traveled by the ship to arrive at the destination) is assigned as the node's value. An edge connecting two nodes in consecutive stages represents the segment between the locations corresponding to the two nodes, and the weight on the edge represents the fuel consumption required to complete the segment.

For each run, the sailing speed optimization problem can be represented in the form of a unique multistage graph. Taking the k th run as an example, as the ship has sailed $k-1$ segments following speed profile $\left(v_1^1, v_2^2, \cdots, v_{k-1}^{k-1}\right)$, the traveled distance is $\sum_{j=1}^{k-1} v_j^j \times t_j^j$, the remaining distance to be traveled is $L^k = L - \sum_{j=1}^{k-1} v_j^j \times t_j^j$, and the remaining maximum allowable traveling time is $T^k = T_{\max} - \sum_{j=1}^{k-1} t_j^j$. The departure time of the ship in the k th run is $T_{\text{start}}^k = T_{\text{start}}^1 + \sum_{j=1}^{k-1} t_j^j$. Then, the number of stages in the k th run can be calculated by $N^k = \lfloor (T^k - (T - \Delta t))/T \rfloor + 2$, where $\Delta t = T_{\text{start}}^k - T_{\text{nearest}}^k$ and T_{nearest}^k is the timestamp of the latest historical meteorological forecast before the departure time of the k th run. As shown in Fig. 4, the sailing time t_j^k on the j th segment in the k th run can be calculated by

$$t_j^k = \begin{cases} T - \Delta t, \ j = k \\ T, j \in \{k+1, \ldots, n-1\} \\ T^k - T \times \left(N^k - 1\right) + \Delta t, j = n \end{cases} \tag{14}$$

which implies that the sailing time on the first segment in the k th run is $T - \Delta t$, that on the final segment is $T^k - T \times \left(N^k - 1\right) + \Delta t$, and that on the j th ($j \in \{k+1, \cdots, n-1\}$) segment is T. If the departure time of the ship in the k th run (T_{start}^k) is the timestamp at which the meteorological forecasts are updated, then $T - \Delta t$ is equal to forecast cycle T; otherwise, it is smaller than T. Stage 1 contains a single node representing the departure point of the k th run, and its value is L^k. Stage N^k also contains a single node, which represents

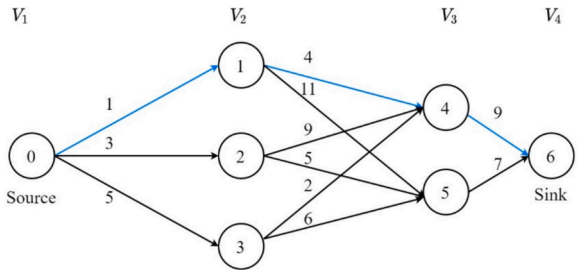


Fig. 3. An example of a four-stage graph.

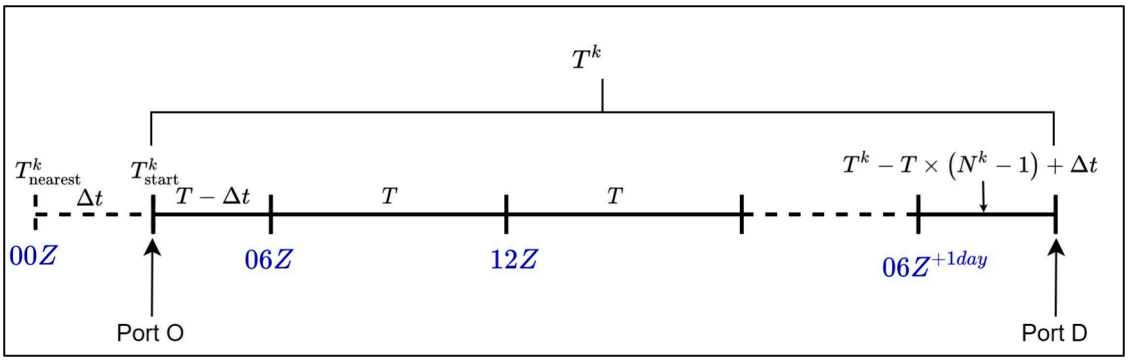


Fig. 4. An illustration of calculating the number of stages in the k th run.

the end of the voyage, and its value is zero, representing that the sink has been reached. To determine the node set of stage $i \in (1, N^k)$, we calculate the range of node values as $[lb, ub]$, where $lb = \max\left(0, L^k - \sum_{j=k}^{k-2+i} v_{\max} \times t_j^k\right)$ and $ub = \max\left(0, L^k - \sum_{j=k}^{k-2+i} v_{\min} \times t_j^k\right)$.[5] If $ub - lb \neq 0$, we discretize the range $[lb, ub]$ based on an interval ζ. The number of discretized values is calculated by $\lfloor (ub - lb)/\zeta \rfloor + 1$, and the discretized values are $lb, lb + 1 \times \zeta, \cdots, ub$, which represent the possible values of all nodes in the i th stage.

Then, we need to establish edges between the nodes in two consecutive stages. Assume that node a in stage i has value $L_i^k(a)$ (i.e., the distance from the current node to the sink node), and node b in stage $i+1$ has value $L_{i+1}^k(b)$. If node a can be connected to node b, this means that the ship can travel from waypoint i to waypoint $i+1$ with distance $L_i^k(a) - L_{i+1}^k(b)$ within the time limit of t_{k+i-1}^k, by sailing within the maximum allowable speed. Therefore, the nodes in stage $i+1$ that can be connected to node a have values in the range of $\left[\max\left(0, L_i^k(a) - v_{\max} \times t_{k+i-1}^k\right), \max\left(0, L_i^k(a) - v_{\min} \times t_{k+i-1}^k\right)\right]$, where $v_{\max} \times t_{k+i-1}^k$ is the maximum allowable sailing distance and $v_{\min} \times t_{k+i-1}^k$ is the minimum allowable sailing distance. By conducting the above process on all nodes in stage i, the nodes in stage i and stage $i+1$ can be connected.

Once the topology of the multistage graph is established, we determine the weights of the edges in the graph. The weight value represents the fuel consumption required for the ship to sail between the locations represented by the two nodes of consecutive stages. Therefore, it is determined based on the sailing time and the predicted FCRs obtained using the ANN model $f^{\text{ANN}}(v, \mathbf{w}, \mathbf{u})$, where v is the sailing speed, $\mathbf{w}$ is the meteorological conditions, and $\mathbf{u}$ is the ship's draught and sailing condition (laden or ballast). To illustrate how to calculate the edge weights in the multistage graph, we take the calculation of the weight on the edge from node a in stage i to node b in stage $i+1$ as an example. First, we need to determine the sailing speed $v_i^k(a \to b)$ to sail from node a to node b based on the distance between the two nodes calculated by their remaining sailing distances. The sailing speed $v_i^k(a \to b)$ is calculated by

$$v_i^k(a \to b) = \frac{L_i^k(a) - L_{i+1}^k(b)}{t_{k+i-1}^k}, \tag{15}$$

where $L_i^k(a)$ is the distance from node a to the sink node, $L_{i+1}^k(b)$ is the distance from node b to the sink node, and t_{k+i-1}^k is the sailing time limit on the segment between nodes a and b, which is calculated by Eq. (14). Then, we need to determine the meteorological

[5] The value of a node is in the interval $[lb, ub]$, which means that the minimum value of the node is zero. The node associated with zero value means that the ship has arrived at the destination. If the value of the node is negative, it means that the current position of the ship is beyond the voyage and is thus unreasonable. Therefore, we restrict the values of lb and ub to be non-negative to ensure that the ship sails within the allowable range.

conditions on the segment between nodes a and b. As mentioned above, the meteorological conditions at the beginning of the segment represent those along the whole segment. Therefore, we need to estimate the geographical coordinates of, and the arrival time at, node a. We assume that the vessel travels along the rhumb line. Therefore, the geographical coordinates of node a can be extrapolated from the rhumb line (Williams, 2013), as shown below:

$$\delta = \frac{d}{R} \tag{16}$$

$$\varphi_2 = \varphi_1 + \delta \times \cos\theta \tag{17}$$

$$\Delta\psi = \ln\left(\frac{\tan\left(\frac{\pi}{4}+\frac{\varphi_2}{2}\right)}{\tan\left(\frac{\pi}{4}+\frac{\varphi_1}{2}\right)}\right) \tag{18}$$

$$q = \begin{cases} \cos\varphi, & \text{the ship sails on constant latitude course } \varphi \\ \dfrac{|\varphi_2-\varphi_1|}{\Delta\psi}, & \text{otherwise} \end{cases} \tag{19}$$

$$\Delta\lambda = \delta \times \frac{\sin\theta}{q} \tag{20}$$

$$\lambda_2 = \lambda_1 + \Delta\lambda, \tag{21}$$

where δ represents the angular distance; $d = L - L_i^k(a)$ represents the distance between node a and the departure port; R is the radius of the earth; λ_2 and φ_2 are the longitude and latitude of node a, respectively; and $\Delta\lambda$ is the difference in longitude between the origin and destination ports. In addition, the arrival time at node a is T_{start}^k if $i = 1$; otherwise, it is $T_{\text{start}}^k + \sum_{j=k}^{k-2+i} t_j^k$. After we obtain the geographical coordinates and the arrival time at node a, we can obtain the meteorological information for the segment between nodes a and b, denoted by $\mathbf{w}(a)$, where the wind speed, wind direction, and 2-meter temperature are derived from the latest meteorological forecast data issued by the NOAA, while other meteorological variables are obtained from the ERA5 reanalysis data. To compute the predicted FCR, we further need to derive other sailing features, including the draught and sailing condition. We find the record in the noon report whose recording time is the closest to the arrival time at node a, and take the draught and sailing condition stated in this record as the draught and sailing condition of the segment between nodes a and b, which are denoted by $\mathbf{u}(a)$. Through the above process, we calculate the weight $w(a \to b)$ of the edge between nodes a and b by

$$w(a \to b) = f^{\text{ANN}}\left(v_i^k(a \to b), \boldsymbol{w}(a), \boldsymbol{u}(a)\right) \times t_{k+i-1}^k. \tag{22}$$

The multistage graph contains some edges connected to the sink node, and the sailing distance on these edges (i.e., the difference between the values of the start and destination nodes of the edge) is less than the minimum allowable sailing distance, which is calculated by multiplying the time limit of these edges by the minimum allowable sailing speed $v_{\min}$. This means that these edges correspond to segments that directly connect to the destination port, and the vessel can arrive at the destination port within the time limit of these edges when sailing at any speed within the interval $[v_{\min}, v_{\max}]$. As the purpose of our sailing speed optimization model is minimizing the total ship fuel consumption, we should assign these edges the minimum fuel consumption resulting from any speed in the interval $[v_{\min}, v_{\max}]$. We discretize the interval $[v_{\min}, v_{\max}]$ to obtain discretized speed values, denoted by $v_{\min}, v_{\min} + 0.1, \ldots, v_{\max}$, and then for each edge we calculate the ship fuel consumption values corresponding to these discretized speeds. The minimum of these fuel consumption values is taken as the fuel consumption for the corresponding edge. For example, if node a in stage i satisfies the above conditions, the fuel consumption value on the edge between node a and sink node is given by

$$min\left(f^{\text{ANN}}(v_{\min}, \boldsymbol{w}(a), \boldsymbol{u}(a)) \times t_{k+i-1}^k, \ldots, f^{\text{ANN}}(v_{\max}, \boldsymbol{w}(a), \boldsymbol{u}(a)) \times t_{k+i-1}^k\right). \tag{23}$$

After the above procedure, the multistage graph that represents the ship speed optimization problem in the k th run is constructed by defining its topology and assigning weights to its edges. Some paths in the multistage graph have a length less than the total length from the origin port to the destination port, meaning that some nodes in the penultimate stage of the multistage graph cannot be connected to the sink node. When a vessel travels along segments represented by such a path, it cannot arrive at the destination port within the maximum allowable sailing time, indicating that such a path is not a feasible solution of the sailing speed profile. Therefore, to reduce the size of the multistage graph and accelerate the calculation of the shortest path, we delete every path whose length is less than the total length (i.e., L). This is done by removing nodes with a zero outdegree[6] on a stage-by-stage basis, starting from the penultimate stage, and we execute this strategy for the multistage graph formulated in each run.

After trimming the multistage graph, the Dijkstra's algorithm (Dijkstra et al., 1959; Hagberg et al., 2008) implemented in the

[6] Outdegree of a node is the number of edges which are going out from the node in a directed graph.

NetworkX Python library 2.5.1[7] is used to find the shortest path between the source and sink nodes. In the k th run, we obtain the shortest path, which has the minimum total fuel consumption. In this path, the nodes reflect the waypoints that the ship passes from the starting position of the k th run to the destination port D, and the speed of the ship navigating on the segment between consecutive nodes can be calculated according to Eq. (14). The shortest path reveals the optimal speed profile for the remaining segments under the currently forecasted meteorological conditions. Recall that the segmentation of routes in Section 5.1 ensures that the sailing time on each path in the multistage graph is no larger than the maximum allowable sailing time. This indicates that the ship must reach the destination port within the specified time, which prevents the ship from always sailing at the minimum permissible speed. The first speed value in the optimal speed profile obtained in the k th run is the optimal speed in the first segment of the remaining voyage in the k th run, which is also the ship's optimal speed in the k th segment of the overall voyage, that is v_k^k. Therefore, we update the topology and weights of the multistage graph in a recursive manner at most n times, and take the first speed value in the speed profile obtained in each run to form the final optimal sailing speed profile, i.e., $\left(v_1^1, \cdots, v_m^m\right)$, where m is the total number of runs and is no larger than n. The pseudo-code of constructing the multi-stage graph in the k th run is presented in Algorithm 2 of Appendix B.

5.3. Evaluation of model performance

To assess the performance of the proposed model, we compare the total fuel consumption under four different speed profiles, i.e., the optimized sailing speeds based on the constantly updated meteorological forecasts (denoted by $v^f = \left\{v_{f:1}^1, \cdots, v_{f:n_f}^{n_f}\right\}$), the constant sailing speed (denoted by $v^c = \left\{v_{c:1}^1, \cdots, v_{c:n_c}^1\right\}$), the optimized sailing speeds based on the meteorological forecasts announced before departure (denoted by $v^s = \left\{v_{s:1}^1, \cdots, v_{s:n_s}^1\right\}$), and the optimized sailing speeds based on the actual meteorological conditions (denoted by $v^a = \left\{v_{a:1}^1, \cdots, v_{a:n_a}^1\right\}$). n_f, n_c, n_s, and n_a are the total numbers of segments in the four cases, respectively. We calculate the overall fuel consumption under the four speed profiles according to the following formulas:

$$FC_{\text{proposed}} = \sum_{i=1}^{n^f} f^{\text{ANN}}\left(v_{f:i}^i, \boldsymbol{w}_{f:i}, \boldsymbol{u}_{f:i}\right) \times t_{f:i}^i \tag{24}$$

$$FC_{\text{constant_speed}} = \sum_{i=1}^{n_c} f^{\text{ANN}}\left(v_{c:i}^1, \boldsymbol{w}_{c:i}, \boldsymbol{u}_{c:i}\right) \times t_{c:i}^1 \tag{25}$$

$$FC_{\text{static}} = \sum_{i=1}^{n_s} f^{\text{ANN}}\left(v_{s:i}^1, \boldsymbol{w}_{s:i}, \boldsymbol{u}_{s:i}\right) \times t_{s:i}^1 \tag{26}$$

$$FC_{\text{actual}} = \sum_{i=1}^{n_a} f^{\text{ANN}}\left(v_{a:i}^1, \boldsymbol{w}_{a:i}, \boldsymbol{u}_{a:i}\right) \times t_{a:i}^1, \tag{27}$$

where $t_{f:i}^i(i = 1, \cdots, n_f)$, $t_{c:i}^1(i = 1, \cdots, n_c)$, $t_{s:i}^1(i = 1, \cdots, n_s)$, and $t_{a:i}^1(i = 1, \cdots, n_a)$ are the sailing times on the i th segment in the four cases, respectively; $\mathbf{w}_{f:i}$, $\mathbf{w}_{c:i}$, $\mathbf{w}_{s:i}$, and $\mathbf{w}_{a:i}$ are the meteorological conditions that the ship encounters on the i th segment in the four cases, respectively, which are retrieved from the ERA5 data; and $\mathbf{u}_{f:i}$, $\mathbf{u}_{c:i}$, $\mathbf{u}_{s:i}$, and $\mathbf{u}_{a:i}$ are the draught and sailing conditions on the i th segment in the four cases, respectively. To obtain a constant sailing speed v^c, we first calculate the lower bound c_{lb} of the constant speed by dividing the sailing distance by the corresponding steaming time, i.e., $c_{lb} = \frac{L}{T_{\max}}$; the upper bound c_{ub} of the constant speed in this study is set as 18.0 knots. The speed interval $[c_{lb}, c_{ub}]$ is discretized in intervals of 0.1, and the total fuel consumption of the ship travelling at each discretized speed value is calculated based on the NOAA meteorological forecast data. The discretized speed value corresponding to the minimum fuel consumption is taken as the constant sailing speed of the ship. To evaluate the performance of the proposed model, we calculate e_1, e_2, and e_3 according to Eqs. (28), (29), and (30), where e_1 evaluates the fuel saving of the proposed model compared with the constant sailing speed scheme; e_2 evaluates the fuel saving of the proposed model compared with the static speed optimization model; and e_3 measures the relative difference between the optimized fuel consumption based on the constantly updated meteorological forecasts and that based on the actual meteorological conditions.

$$e_1 = \frac{FC_{\text{proposed}} - FC_{\text{constant_speed}}}{FC_{\text{constant_speed}}} \times 100\% \tag{28}$$

$$e_2 = \frac{FC_{\text{proposed}} - FC_{\text{static}}}{FC_{\text{static}}} \times 100\% \tag{29}$$

[7] https://networkx.org/.

$$e_3 = \left| \frac{FC_{\text{proposed}} - FC_{\text{actual}}}{FC_{\text{actual}}} \times 100\% \right|. \tag{30}$$

6. Computational experiments

6.1. Prediction of ship fuel consumption rates

The MLPRegressor function in the scikit-learn library 0.24.2[8] is employed to develop the ANN model for predicting the ship FCRs. The hyperparameters of the ANN regressor are found by applying the grid search method with five-fold cross validation to the training set. Table 3 shows the interpretations, search space, and optimal values of the hyperparameters, namely hidden_layer_size, activation function, solver, and max_iter. The search space of hidden_layer_size is determined by the number of hidden layers, which takes a value in $\{1, 2, 3, 4, 5, 6, 7\}$, and the number of neurons in each hidden layer, which takes a value in $\{8, 16, 32, 64\}$. In addition, the structure of the hidden layers is constrained such that the middle hidden layer has the largest number of neurons, and the number of neurons in the left and right hidden layers follows the rule that the number of neurons in the $(i-1)$ th hidden layer is less than or equal to that in the i th hidden layer on the left, and larger than or equal to that in the i th hidden layer on the right. After constructing the ANN model using the entire training set with the optimal hyperparameters, the model performance on the test set is as shown in Table 4. The MSE, MAE, and MAPE of the ANN regressor are 0.016, 0.081, and 5.95 %, respectively.

6.2. Validation of ship sailing speed optimization model

We examine two voyages recorded in the test set, denoted voyages I and II. Voyage I is an 11-day laden voyage, covering an overall sailing time of 261 h and an overall sailing distance of 2701.56 nm. Voyage II is an 11-day laden voyage, covering an overall sailing time of 266 h and an overall sailing distance of 3584.99 nm. Detailed information on voyages I and II, including the average sailing speed, draught, and sailing conditions and the surrounding meteorological conditions derived from ERA5 data, is presented in Appendix A.

We assess the effectiveness of the proposed sailing speed optimization model on voyages I and II. The computational experiments are conducted on a laptop computer (Intel Core i7-12700H CPU, 2.30 GHz, memory 40.0 G). We discretize the range of the remaining distance to the destination port, represented by the nodes in each stage in the multistage graph, in intervals ζ of 1 nm. It takes 146 min and 220 min to obtain the optimal speeds for voyages I and II, respectively. We also explore the complexity of the proposed method by applying it to the problem of different size, as shown in Appendix C. The optimized sailing speed and fuel consumption in each segment of voyages I and II are shown in Fig. 5(a) and Fig. 5(b), respectively. The fuel consumption in each segment is calculated by multiplying the steaming time by the hourly FCR under the actual meteorological conditions. This is achieved by inputting the optimized sailing speed, vessel operational data, and actual meteorological data provided by the ERA5 data into the ANN model. It can be seen that the fuel consumption in each segment increases as the ship sailing speed increases. After speed optimization, voyage I is divided into 39 segments. The ship sails for 6 h in each of the first 38 segments and for 0.97 h in the last segment, with an overall sailing time of 228.97 h for the voyage. Voyage II is divided into 45 segments. The ship sails for 4 h in the first and last segments and for 6 h in all other segments, and thus the total sailing time is 266 h. These results indicate that the vessel can arrive at the destination within the specified period when sailing at the optimized speeds on voyages I and II.

Table 5 shows the total fuel consumptions under four different speed profiles, i.e., the optimized sailing speeds based on the constantly updated meteorological forecasts (FC_{proposed}), the constant sailing speeds ($FC_{\text{constant_speed}}$), the optimized static sailing speeds (FC_{static}), and the optimized sailing speeds based on the actual meteorological conditions (FC_{actual}). It can be seen that the ship consumes 323.01 mt of fuel to fulfill voyage I and 455.91 mt of fuel to fulfill voyage II, while ensuring the arrival time at the destination port. Compared with the total fuel consumption predicted from the speed profiles reported in the noon reports, i.e., $FC_{\text{predicted}}$ in Table 5, the optimized sailing speed profile reduces fuel consumption by 9.10 % and 0.13 % on voyages I and II, respectively. Additionally, compared with the constant sailing speed scheme, the proposed model reduces the total fuel consumption by 9.46 % and 2.21 % for voyages I and II, respectively, while the corresponding reductions relative to the static speed optimization model are 13.50 % and 2.41 %. Over the two voyages, the proposed method achieves significant reductions in fuel consumption by 5.35 % compared with a constant sailing speed scheme and by 7.34 % compared with a static speed optimization model.

Furthermore, we measure the relative difference between the total fuel consumption when the vessel travels at optimized speeds based on constantly updated meteorological forecasts and when it sails at optimized speeds based on actual meteorological conditions. For voyage I, this relative difference is 4.10 %, and for voyage II, it is 15.34 %, indicating that speed optimization based on constantly updated meteorological forecasts performs similarly to speed optimization based on actual meteorological conditions. Therefore, the speed optimization method proposed in this study can effectively reduce fuel consumption even when only meteorological forecasts are available and the actual conditions are unknown. Therefore, it can be efficiently used to plan the sailing speed profiles of a company's next voyages.

[8] https://scikit-learn.org/.

Table 3
Hyperparameters in the ANN regressor.

Hyperparameter	Interpretation	Search space	Best value
hidden_layer_size	A list representing the number of neurons in each hidden layer, where the i th element is the number of neurons in the i th hidden layer.	The number of hidden layers:$\{1, 2, 3, 4, 5, 6, 7\}$ The number of neurons in each hidden layer:$\{8, 16, 32, 64\}$	(16, 32, 32, 32, 16, 8, 8)
activation function	Activation function employed in the hidden layer	{tanh, ReLU, sigmoid}	ReLU
solver	The solver for weight optimization.	{lbfgs, sgd, adam}	Adam
max_iter	Maximum number of iterations.	$\{1000, 2000, 3000, 4000, 5000, 6000, 7000\}$	6000

Table 4
Predictive performance of the ANN regressor.

Model	MSE	MAE	MAPE
ANN regressor	0.016	0.081	5.95 %

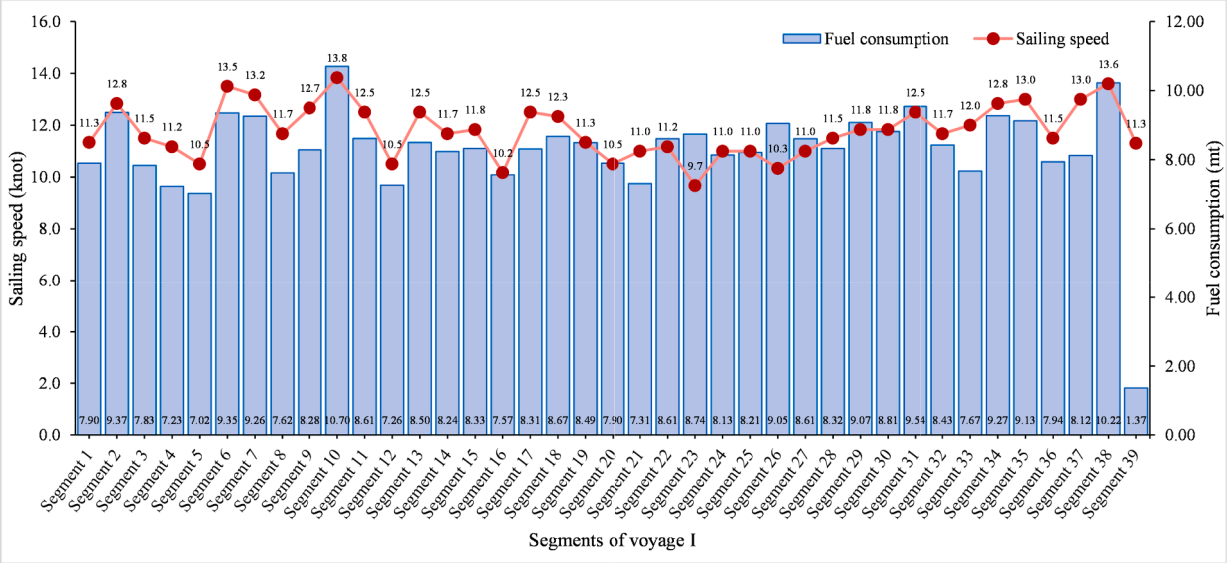
7. Conclusion and future research

The shipping industry's reliance on fossil fuels has led to significant GHG emissions, posing environmental challenges. The IMO has implemented initiatives aimed at reducing emissions from shipping activities, passing regulations and requirements with which shipping companies must comply. Sailing speed optimization has emerged as a cost-effective strategy to improve ship energy efficiency, thus reducing GHG emissions. In previous studies, the meteorological conditions used for sailing speed optimization were obtained before ship departure, and predictions of the ship FCRs based on such meteorological data were used to plan the ships' speed profiles. However, these static meteorological conditions do not reflect the ever-changing marine environment. Consequently, in those studies, the optimized sailing speeds were unable to maximize fuel savings. To address this issue, this study develops a vessel sailing speed optimization model considering dynamic meteorological conditions, and validates the effectiveness of the proposed model based on a two-phase framework. Specifically, in the first phase, based on a dataset incorporating noon reports and ERA5 meteorological information, an ANN model is constructed to predict the vessel's FCRs under various meteorological conditions. In the second phase, a sailing speed optimization model is constructed based on the fuel consumption predicted by the ANN model. This model constantly optimizes the vessel's sailing speed profile to minimize the total fuel consumption over the remaining voyage once the newly updated meteorological forecasts provided by the NOAA are obtained.

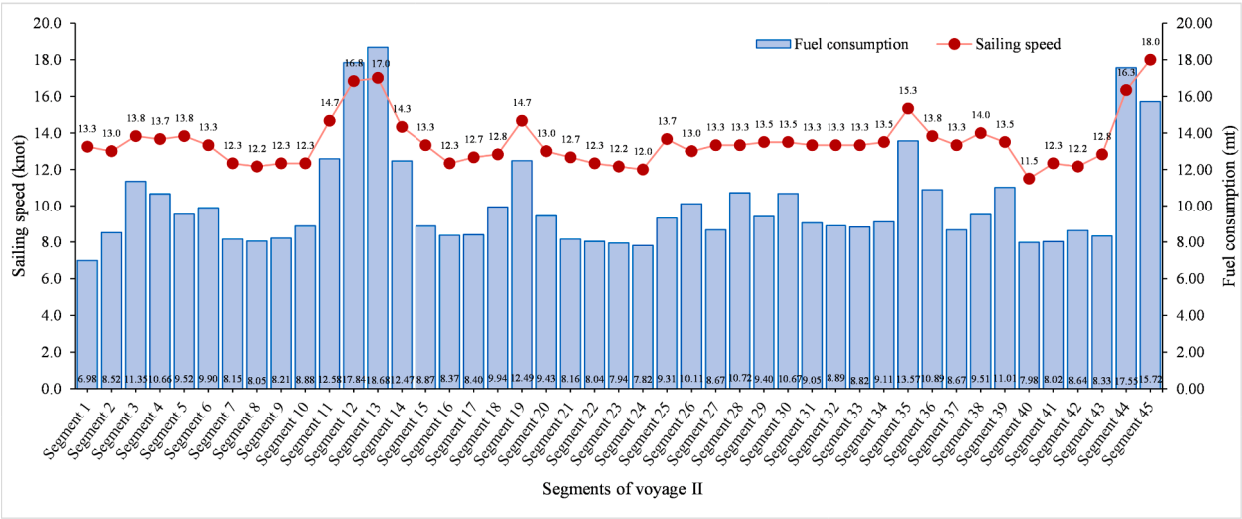
In the numerical experiments, two 11-day voyages are chosen to validate the effectiveness of the proposed sailing speed optimization model. On voyages I and II, the model achieves reductions of 9.10 % and 0.13 % in fuel consumption, respectively, compared with the total fuel consumption calculated from the speed profiles stated in the noon reports. The performance of the proposed model is also compared with those of two other speed decision schemes. Compared with a constant sailing speed scheme, the proposed model reduces the total fuel consumption by 9.46 % and 2.21 % for voyages I and II, respectively. Furthermore, compared with a static speed optimization model, the proposed model reduces the total fuel consumption by 13.50 % and 2.41 % for voyages I and II, respectively. In addition, we measure the relative difference between the total fuel consumptions of two speed configurations: one optimized using meteorological forecasts and the other optimized using actual meteorological conditions. For voyage I, this relative difference is 4.10 %, and for voyage II, it is 15.34 %, indicating that the speed optimization method proposed in this study can effectively reduce fuel consumption in the absence of knowledge of the actual meteorological conditions.

This study's contributions are twofold. First, the study proposes a sailing speed optimization model that considers dynamic meteorological conditions at sea. The proposed model is innovative in that it re-optimizes the ship's sailing speeds for the remaining voyage based on the most recently updated meteorological forecasts, ensuring that the speed optimization results remain accurate and reliable despite changing maritime environmental conditions. In this model, the speed optimization problem is represented as a multistage graph. When a new meteorological forecast is generated, we update the topology of the multistage graph by determining the ship's new starting position, and adopt the ANN model to update the weights of the graph based on the meteorological conditions provided by the latest meteorological forecast. The shortest path, corresponding to the minimum total fuel consumption over the voyage, is found by utilizing the Dijkstra's algorithm, ultimately yielding the optimal speed profile under the currently forecasted meteorological conditions. This updating process is repeated to generate the optimal sailing profile for the remaining voyage segments until the ship arrives at the destination port. The model is validated on two voyages, achieving significant fuel savings compared with two alternative speed decision schemes, i.e., a constant sailing speed scheme and a static speed optimization model. In addition, the speed optimization method proposed in this study can effectively reduce fuel consumption even in the absence of knowledge of the actual meteorological conditions. Second, regardless of the ship type and route, the proposed model could be integrated with existing ship energy management systems, providing an additional layer of ship sailing behavior optimization to further improve energy efficiency. The model's flexibility and adaptability will make it a valuable tool for the shipping industry in terms of achieving sustainable growth and reducing GHG emissions.

This study also has several limitations, which point out directions for future research. First, we only employ Dijkstra's algorithm for determining the shortest path of the multi-stage graph due to its widespread usage in the field of maritime transportation. For example,



(a) Voyage I



(b) Voyage II

Fig. 5. Fuel consumption and sailing speed after optimization on the two selected voyages. (a) Voyage I. (b) Voyage II.

Table 5
The performance of the proposed sailing speed optimization model on voyages I and II.

Voyage	FC_{proposed}(mt)	$FC_{\text{predicted}}$(mt)	$FC_{\text{constant_speed}}$(mt)	FC_{static}(mt)	FC_{actual}(mt)	e_1	e_2	e_3
Voyage I	323.01	355.33	356.75	373.43	310.29	−9.46 %	−13.50 %	4.10 %
Voyage II	455.91	456.52	466.23	467.19	395.27	−2.21 %	−2.41 %	15.34 %

Note: $e_1 = \frac{FC_{\text{proposed}} - FC_{\text{constant_speed}}}{FC_{\text{constant_speed}}} \times 100\%$; $e_2 = \frac{FC_{\text{proposed}} - FC_{\text{static}}}{FC_{\text{static}}} \times 100\%$; $e_3 = \left|\frac{FC_{\text{proposed}} - FC_{\text{actual}}}{FC_{\text{actual}}} \times 100\%\right|$.

Silveira et al. (2019) adopt Dijkstra's algorithm to identify a potential safe route in a graph constructed based on the historical Automatic Identification System data, where nodes are cells of a grid covering the geographical area being studied and the weights of directional edges are inversely related to ship movements between cells. Wang et al. (2019) generate a three-dimensional weighted

graph where nodes represent the waypoints and edges represent the cost, i.e., fuel consumption. Other methods, such as A* algorithm and dynamic programming, are also commonly employed for finding the shortest path in a graph. Future research can involve comparing various algorithms to identify the most efficient one for finding the shortest path in a multi-stage graph.

In addition, we ignore the impact of fuel consumption prediction uncertainty caused by inaccurate meteorological forecasts on the speed optimization results. In this paper, the optimization model proposed utilizes continuously updated meteorological forecasts to capture the continuously changing marine environment. By leveraging the latest meteorological forecast, the optimization model constantly optimizes the sailing speed for the remaining voyage until the ship arrives at the destination port. This iterative process ensures that the optimized sailing speed is tailored to the most up-to-date meteorological conditions, effectively reducing fuel consumption and costs. The meteorological forecasts employed in this study are updated four times daily, with each update issuing a single forecast. This type of forecast is known as deterministic meteorological forecast. Another type of meteorological forecast data is known as ensemble forecasts, which simulate multiple meteorological scenarios by adding small perturbations to the initial weather values. This ensemble forecast provides multiple meteorological forecast values, better capturing the uncertainty associated with weather forecasts. Future research endeavors can focus on constructing a novel sailing speed optimization model based on this continuously updated ensemble forecast data. In this new model, we can estimate the probability distribution of fuel consumption by considering multiple weather members within the ensemble forecast data. An advanced ship speed optimization model can be constructed by further considering the uncertainty of ship fuel consumption.

CRediT authorship contribution statement

Xi Luo: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Ran Yan:** Writing – review & editing, Supervision, Project administration, Methodology, Funding acquisition, Conceptualization. **Shuaian Wang:** Writing – review & editing, Supervision, Methodology, Funding acquisition, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

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Appendix A. Original ship sailing features for the two voyages selected as the test set

Table A1
Original vessel sailing conditions for voyage I.

Date	Fuel consumption rate (mt/h)	Average sailing speed (knot)	Draught (m)	Sailing condition	Relative wind direction (degree)	Wind speed (m/s)	Mean wave direction (degree)	Mean wave period (s)	Significant height of combined wind waves and swell (m)	Forecast surface roughness (m)	2 m temperature (°C)
2020/7/14	1.3478	10.13	17.5	1	164.2271	8.9022	7.1505	5.6554	1.5409	2.99E-04	20.3210
2020/7/15	1.3542	9.54	17.5	1	156.5666	7.6821	158.3993	6.3514	1.6400	3.08E-04	21.3732
2020/7/16	1.3542	9.33	17.5	1	138.2798	3.9703	43.9713	7.9327	1.6830	1.26E-03	20.8831
2020/7/17	1.3542	10.54	17.5	1	35.2410	3.0905	67.9185	6.6059	0.9274	1.74E-01	22.8000
2020/7/18	1.3478	10.96	17.5	1	169.7410	4.5898	3.7152	4.3399	0.7082	5.94E-02	23.4940
2020/7/19	1.3542	10.58	17.5	1	175.2500	5.6923	14.5434	3.9854	0.6232	5.99E-05	23.5432
2020/7/20	1.3542	10.71	17.5	1	178.1069	5.9768	21.2735	3.4014	0.5661	2.39E-02	24.8661
2020/7/21	1.3542	10.83	17.5	1	3.3383	1.1273	63.0173	3.0068	0.2739	4.32E-02	24.6442
2020/7/22	1.3478	10.87	17.5	1	59.4343	5.3178	169.4696	3.5963	0.6087	5.65E-05	25.9084
2020/7/23	1.3542	10.71	17.5	1	65.8526	6.0317	146.0129	4.0811	0.7314	5.25E-03	25.6781
2020/7/24	1.3542	10.67	17.5	1	52.1370	7.4369	131.3939	4.4977	1.0341	4.86E-02	24.7836
2020/7/25	1.3478	10.74	17.5	1	6.0543	5.9413	169.8358	4.4851	0.8542	6.11E-05	27.1862
Total sailing time		261.0 h									
Total sailing distance		2701.56 nm									
Total fuel consumption		352.98 mt									

Table A2
Original vessel sailing conditions for voyage II.

Date	Fuel consumption rates (mt/h)	Average sailing speed (knot)	Draught (m)	Sailing condition	Relative wind direction (degree)	Wind speed (m/s)	Mean wave direction (degree)	Mean wave period (s)	Significant height of combined wind waves and swell (m)	Forecast surface roughness (m)	2 m temperature (°C)
2019/2/13	1.3542	10.49	17	1	118.4914	6.6456	104.7804	7.1267	1.6089	4.21E-02	28.4665
2019/2/14	1.3542	10.92	17	1	71.6087	7.5666	125.4316	7.1884	1.9317	2.13E-02	28.1699
2019/2/15	1.3542	10.92	17	1	67.4272	9.4594	114.3748	6.6539	1.9712	1.00E-03	27.9721
2019/2/16	1.3542	11.54	17	1	175.5482	6.9029	1.1794	5.4694	1.2431	4.02E-03	28.1760
2019/2/17	1.3542	10.04	17	1	169.3271	8.1297	0.1800	5.3415	1.1895	4.41E-02	27.6263
2019/2/18	1.3750	9.83	17	1	175.8656	7.1494	166.0797	6.5738	1.5390	1.14E-04	27.3241
2019/2/19	1.3540	10.80	17	1	135.4831	5.3609	95.2685	8.0560	1.7688	4.54E-05	27.6568
2019/2/20	1.3542	11.75	17	1	133.2608	4.0265	105.3969	8.4975	1.7343	2.46E-05	27.2466
2019/2/21	1.3542	12.08	17	1	103.2293	3.9204	109.9028	8.4649	1.7177	3.02E-05	27.0003
2019/2/22	1.3750	11.04	17	1	87.2013	6.3779	119.9324	7.6561	1.7526	8.04E-05	26.4363
2019/2/23	1.3540	10.80	17	1	110.2064	1.5615	149.4959	4.8875	0.6133	7.49E-03	27.0450
2019/2/24	1.3667	11.50	17	1	67.9050	4.6753	109.4991	3.9811	0.6849	1.26E-02	26.3614
Total sailing time		266.0 h									
Total sailing distance		3584.99 nm									
Total fuel consumption		361.24 mt									

Appendix B. Pseudo-code of the sailing speed optimization model

Algorithm 1. Sailing speed optimization

Input: The total length L from the origin port to the destination port; The maximum allowable travelling time $T_{\max}$; The forecast cycle T; The departure time T^1_{start}; Timestamp T^1_{nearest} of the latest historical meteorological forecast before the departure time of the first run; The minimum allowable sailing speed $v_{\min}$; The maximum allowable sailing speed $v_{\max}$; Distance interval ζ.

Output: The speed profile $(v^1_1, v^2_2, \cdots, v^m_m)$.

For $k = 1, \cdots, n$ **do**:
  // The ship has sailed $k-1$ segment with speed profile $\mathscr{V} = (v^1_1, \cdots, v^{k-1}_{k-1})$ and travel time $\mathscr{T} = (t^1_1, \cdots, t^{k-1}_{k-1})$
  Calculate the remaining distance to be travelled in the k th run: $L^k = L - \sum_{j=1}^{k-1} v^j_j \times t^j_j$
  Calculate the remaining maximum allowable traveling time in the k th run as $T^k = T_{\max} - \sum_{j=1}^{k-1} t^j_j$
  Calculate the departure time in the k th run as $T^k_{\text{start}} = T^1_{\text{start}} + \sum_{j=1}^{k-1} t^j_j$
  Calculate the number of stages: $N^k = \lfloor (T^k - (T - \Delta t))/T \rfloor + 2$, where $\Delta t = T^k_{\text{start}} - T^k_{\text{nearest}}$ and T^k_{nearest} is the timestamp of the latest historical meteorological forecast before the departure time of the k th run
  Calculate the sailing time t^k_j on the j th ($j = k, k+1, \cdots, N^k - 1$) segment in the k th run ← Eq. (14)
  Construct the multistage graph ← Call **Algorithm 2**
  Obtain the optimal path $\left(x^k_k, x^k_{k+1}, \cdots, x^k_{k+N^k-1}\right)$ ← Dijkstra's algorithm on G_k, where x^k_{k+j} is the node in stage $k+j$ of G_k
  Obtain the optimal speed at the k th run $v^k_k = (x^k_{k+1} - x^k_k)/t^k_k$, i.e., the speed of the ship in the k th segment
  Update speed profile $\mathscr{V} \leftarrow (v^1_1, \cdots, v^{k-1}_{k-1}, v^k_k)$ and travel time $\mathscr{T} \leftarrow (t^1_1, \cdots, t^{k-1}_{k-1}, t^k_k)$
  If $\sum_{j=1}^{k} v^j_j \times t^j_j \geq L$, **then**
    $m = k$
    Return $(v^1_1, v^2_2 \cdots, v^m_m)$
  End if
End

Algorithm 2. The construction of multi-stage graph

Input: The minimum allowable sailing speed $v_{\min}$; The maximum allowable sailing speed $v_{\max}$; The remaining distance L^k to be travelled in the k th run; The number of stages N^k in the k th run; The sailing time t^k_j on the j th ($j = k, k+1, \cdots, n$) segment in the k th run

Output: The multi-stage graph G_k

Initialize a multi-stage graph G_k in the k th run
Determine the node set Q_i of each stage:

$$Q_i = \begin{cases} (L^k), i = 1 \\ Q^*_i, i \in [2, N^k - 1] \\ (0_i), i = N^k \end{cases}, \text{ //}0_i \text{ indicates the node with a value of zero on the } i \text{ th stage}$$

where

$$Q^*_i = \begin{cases} (lb, lb + 1 \times \zeta, \cdots, ub), ub \neq lb = 0 \\ (0), ub - lb = 0 \end{cases},$$

$$lb = \max\left(0, L^k - \sum_{j=k}^{k-2+i} v_{\max} \times t^k_j\right) \text{ and } ub = \max\left(0, L^k - \sum_{j=k}^{k-2+i} v_{\min} \times t^k_j\right).$$

Add node set of each stage into the graph G_k
For each node a in $Q_i (i = 1, \cdots, N^k - 1)$ **do**:
  If $a > v_{\min} \times t^k_{k+i-1}$
    For each node b in Q_{i+1} **do**:
      If $b \in \left[\max\left(0, L^k_i(a) - v_{\max} \times t^k_{k+i-1}\right), \max\left(0, L^k_i(a) - v_{\max} \times t^k_{k+i-1}\right)\right]$, **then**
        $G_k \leftarrow (a, b)$// Add edge (a, b) to G_k
        Calculate the weight of edge (a, b): $w(a \rightarrow b)$ ← Eq. (22)
      End if
    End for
  Else
    $G_k \leftarrow (a, 0_{i+1})$, where $0_{i+1} \in Q_{i+1}$ // Add edge $(a, 0_{i+1})$ to G_k
    Calculate the weight of edge $(a, 0_{i+1})$: $w(a \rightarrow 0_{i+1})$ ← Eq. (23)
  End if
End for
// Remove nodes with zero outdegree on a stage-by-stage basis
For $i = 1, \cdots, N^k - 1$ **do**:
  $G_k \leftarrow G_k / \{x | x \in Q_{N^k - i}, \text{outdegree}(x) = 0\}$
End for
Return the multi-stage graph G_k

Appendix C. The complexity of the sailing speed optimization model

The proposed sailing speed optimization model is applied to problems of different sizes. By adjusting the total sailing distance and distance interval ζ, multistage graphs with different numbers of nodes and edges are generated in the first run, as illustrated in Table C.1. The time required to obtain the optimal speed using the proposed model in the first run is shown in Table C.1.

Table C1
The running time for problems with different sizes.

Total sailing Distance (nm)	Distance interval ζ (nm)	The number of nodes	Running time (minute)
3584.0	1.0	34,621	18.86
3584.0	1.5	23,096	7.32
3584.0	2.0	17,327	5.11
3584.0	5.0	6,834	1.40
3584.0	10.0	3,320	0.98
2368.6	1.0	25,198	19.17
2101.0	1.0	19,756	6.68
1836.0	1.0	15,153	4.97
1545.0	1.0	11,054	10.32
1240.5	1.0	7,620	2.60
529.3	1.0	1,223	0.57

The main computational load of the proposed method is contributed by finding the shortest path on the multistage graph. In this paper, we employ the Dijkstra's algorithm implemented by the NetworkX python library[9] to find the shortest path. The Dijkstra's algorithm has the complexity of $O(n^2)$, where n represents the number of nodes in a graph. The Figure C.1 shows the computational time against the square of the number of nodes in different settings. It can be observed that the computational time demonstrates a linear relationship with the square of the number of nodes. The R^2 for the linear fitting is 0.72, which shows a good fit. Therefore, the complexity of the proposed method is $O(n^2)$.

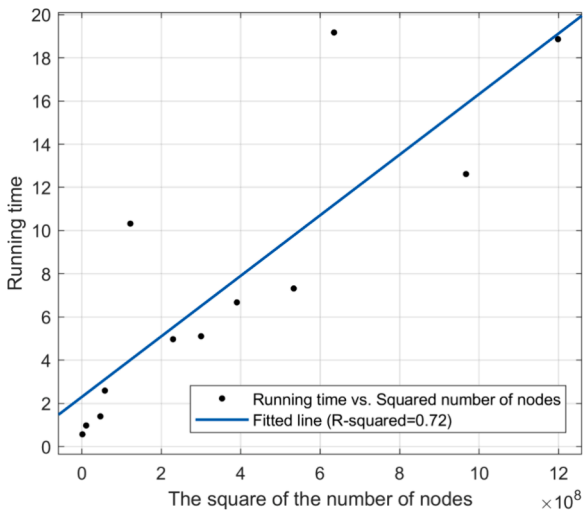

Fig. C1. The fitted line between the running time and the square of the number of nodes.

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